Safety, Quality, and Applicability of Large Language Model–Generated Exercise Prescriptions: A Systematic Review

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Author-review draft | 28 October 2026. The completed synthesis remains 35 studies (25 Class A, 7 Class B, and 3 Class C) reported in 37 reports; W19 is the documented full-text exclusion. The 755-row title/abstract ledger records 533 exclusions and 222 reports sought for full text. A supplementary review of the 189 previously unlinked reports classified 168 as abstract-level exclusion recommendations, 14 as having completed full-text initial assessment, and 7 as not obtained. Of the 14 assessed reports, 8 were recommended for inclusion, 4 for exclusion, and 2 remained undecided. These are AI-assisted recommendations or unresolved states, not reconstructed historical researcher decisions. Because the eight new inclusion candidates do not have complete record-level data extraction and quality appraisal in the available workspace, they are not incorporated into the 35-study synthesis. The 303 saved appraisal judgments for the 35-study set remain unchanged. The PRISMA pathway is therefore an updated working pathway rather than a closed final flow.

Abstract

Background: LLMs can rapidly generate exercise plans, but output quality, safety, and real-world applicability represent distinct levels of evidence and require separate assessment.

Objective: To evaluate the quality, adherence to exercise-prescription principles, safety-related content, output stability, and applicability of exercise prescriptions generated by large language models (LLMs).

Methods: Searches yielded 449 main-search records and 597 PubMed update records (1,046 total). After reconciliation of historical duplicate flags, 755 records entered the title/abstract ledger; 533 were excluded and 222 were sought for full text. A supplementary review of 189 previously unlinked reports produced 168 abstract-level exclusion recommendations, 14 completed full-text initial assessments (8 inclusion recommendations, 4 exclusion recommendations, and 2 undecided), and 7 reports not obtained. Recommendation status was kept separate from formal researcher confirmation. The completed synthesis therefore remains 35 studies in 37 reports (A=25, B=7, C=3), with W19 excluded; eight additional inclusion candidates await complete extraction and design-appropriate appraisal. Quality appraisal for the completed synthesis comprises 288 items in a review-specific, unvalidated Class A/B framework and 15 RoB 2 domains for three randomized Class C studies. No independent second reviewer was recorded.

Results: The current author-designated synthesis set comprises 35 studies reported in 37 reports, including simulated cases, repeated generation across models, evaluations using real patient or user data, and three human implementation studies. LLMs generally generated structured plans containing an exercise type and at least one dose dimension, but dose suitability, individualization, risk screening, prompt transparency, and repeated-generation stability varied. W13 evaluated 360 outputs from three models across six clinical scenarios. Claude Sonnet 4.6 contributed to both generation and output classification, raising a potential self-evaluation or measurement-bias concern. Safety outcomes in Classes A and B primarily concerned warnings, contraindications, dose, or expert judgments and should not be interpreted as human adverse events. The three human implementation studies did not adequately report adverse-event outcomes.

Conclusions: The completed 35-study synthesis indicates that LLMs can generate structured exercise-prescription drafts, while completeness, dose suitability, individualization, output safety, reproducibility,

and human safety remain distinct evidence domains. The evidence does not support unsupervised prescribing. The supplementary review has narrowed the 189-report linkage gap, but eight inclusion candidates still require complete extraction and quality appraisal, two full-text assessments remain undecided, and seven reports were not obtained. The PRISMA pathway should not be reported as final until those record-level decisions and W34–W36 pathway corrections are formally documented.

Keywords: large language models; exercise prescription; systematic review; safety; output quality; reproducibility

1. Introduction

Exercise prescription translates health status, physical capacity, and personal goals into actionable activities and doses. Frequency, intensity, time, type, volume, and progression form the FITT-VP framework, but these components must be adjusted in response to the individual [4]. The same training format may impose different demands on a healthy young adult, a person with a chronic condition, or someone with marked muscle weakness. Prescription evaluation should therefore assess not only whether components are present, but also whether the dose is appropriate, monitoring is feasible, and stopping or referral criteria are specified.

LLMs can interpret natural-language requests, organize training schedules, and revise activities and doses through conversation. Primary studies have examined synthetic cases, resistance and running programs, plans generated from real user information, and implementation in home and fitness settings [5,6,10,12,16]. This approach may reduce the effort required to draft plans and explain information, while making prompt design, input completeness, and professional review important determinants of prescription quality. A fluent, well-formatted plan may not make dose or applicability limitations apparent to users.

The evidence assessed in existing evaluations is not uniform. Expert ratings assess output content; text-similarity measures assess linguistic or semantic proximity; user questionnaires reflect experience; and changes in physical activity or fitness concern human implementation. Even when all are described as “safety,” guideline adherence, expert judgments of risk, and actual adverse events are not interchangeable. Without these distinctions, generatability, acceptability, or reproducibility may be mistaken for clinical effectiveness or safety.

This review focuses on studies in which an LLM actually generated or materially modified an exercise prescription. It examines output quality, adherence to prescription principles, safety-related concerns, stability, and applicability. Evidence from standardized or simulated outputs, outputs based on real-world data, and human implementation is synthesized separately to clarify what each evidence class can support.

2. Methods

2.1 Protocol and reporting framework

Study selection and searches were organized with reference to PRISMA 2020 and PRISMA-S [1,2]. The current review follows an independently revised protocol (Pathway B). The original PROSPERO record, CRD420261495731, is not presented as prospective registration of the current protocol. The available revision record is dated 11 October 2026; this date is not taken to mean that the revised rules were established before searching or screening. The protocol change is reported transparently in the Methods and limitations.

Evidence was stratified into Classes A, B, and C. Classification was based on data source, the LLM’s role, and whether exercise was implemented, rather than on whether experts participated in evaluation. PlanFitting (W29) generated plans from real user information without continuous implementation and was classified as

Class B. The inclusion body myositis study (W31) evaluated standardized clinical questions and was classified as Class A. Results across the three classes were not combined into an overall effect.

2.2 Eligibility criteria and evidence classes

Studies were eligible when they reported extractable empirical results and an LLM actually generated or materially modified an exercise prescription. A prescription had to specify an exercise type or activity and at least one actionable dose dimension: frequency, intensity, time, volume, or progression. Health education, explanations of plans written by humans, general conversation, or stand-alone risk/referral advice without the required prescription content did not meet the core definition.

Class A comprised evaluations of generated prescriptions in standardized, simulated, or synthetic cases. Class B comprised evaluations of outputs generated from real user or patient data. Class C required that the generated plan be implemented in people and that implementation or human outcomes be extractable. User interaction with a Class B system did not constitute exercise implementation. For Class C studies in which the independent causal contribution of the LLM could not be identified, the overall intervention could still be described, but its effects were not attributed to the model.

Both public and proprietary LLMs were eligible; no language or study-design restrictions were prespecified. Conference full papers and preprints could be included when they contained complete Methods and Results and duplicate reports had been checked. Full-text availability was recorded separately from eligibility. Study-level eligibility was based on the full-text generation process, the presence of an activity/type and an actionable dose dimension, and independent empirical evaluation.

2.3 Information sources and search records

Existing search materials from PubMed/MEDLINE, Embase, Scopus, Web of Science, IEEE Xplore, and arXiv were reused; no new searches were conducted. The historical main search yielded 449 records and the PubMed update yielded 597, for 1,046 records in total. The original paper records the main search as having been run on 27–28 September 2026 and the update on 5 October 2026. These dates are available only as historical entries. Original documentation of the exact search strategy, date, fields, and limits is incomplete for individual databases, so these details have not all been independently verified.

Original exports and historical record-level data were retained separately. The recovered PubMed main-search text contains 68 PMIDs, but this does not mean that original exports are available for all 449 records from the six sources. The current IEEE file contains 18 RIS records, a subset of the 27 records recorded in the historical workbook. Existing mappings support parts of the deduplication audit but do not replace missing complete database search documentation.

2.4 Deduplication and study selection

The 1,046 original records retain source identifiers and historical deduplication mappings. The source workbook flags 295 records as duplicates; four were restored as distinct reports, leaving 291 net removals and a 755-row report-level screening inventory. The current ledger contains 533 exclusions and 222 reports sought for full text. Of the 533 exclusions, 530 were recovered from the saved log and three (MAIN-0271, MAIN-0282, and MAIN-0290) were confirmed on 27 October 2026; those three decisions were not backdated. Historical labels, AI advice, saved screening outcomes, and supplementary-review recommendations were retained as separate evidence fields.

The 222 reports sought for full text comprise 33 exact DOI links to the existing candidate inventory and a 189-report supplementary-review queue. The supplementary review classified 168 records as abstract-level exclusion recommendations, 14 reports as having completed full-text initial assessment, and 7 reports as not obtained. Among the 14 assessed reports, 8 were recommended for inclusion, 4 for

exclusion, and 2 remained undecided. These outcomes are not treated as historical human decisions. Record-level files identifying the 14 assessed reports, the eight new inclusion candidates, and the seven unobtained reports were not present in the available workspace, so the eight candidates were not added to the completed synthesis or quality appraisal. MAIN-0076 retains a provisional author preference toward exclusion rather than a final decision. W34–W36 retain a cross-stage discrepancy between saved title/abstract labels and the current candidate set. W13 and W29 each have an associated report version and count once at study level.

2.5 Data extraction and source traceability

Data were extracted by Study_ID and Report_ID, including author, year, setting, design, case or population, model and version, prompt conditions, LLM role, activities and doses, evaluation method, outcomes, safety, principal findings, and limitations. The 35 studies were classified as A (n=25), B (n=7), or C (n=3); multiple reports for W13 and W29 were linked to one study each.

Information not reported in a source that was read was coded as NR. Country was not inferred from author affiliation, and an interface or product name was not automatically treated as the underlying model architecture. Numbers of cases, outputs, users, evaluators, and rating records were extracted separately and were not added into a single sample size.

2.6 Outcomes and safety definitions

The review focused primarily on prescription quality, completeness, accuracy, individual fit, and safety-related content. Repeated generation, model differences, user acceptance, implementation feasibility, and human physical activity or fitness were considered within their respective evidence classes. Safety was separated into output-level evaluation and events observed during human implementation. Output-level assessment included contraindications, risk warnings, dose, monitoring, stopping/referral advice, and expert judgments. Human safety required observations in people and was not inferred from output assessment.

Where prescription-level coding was sufficiently detailed, the “proportion of prescriptions with at least one safety failure” was defined as the number of evaluable prescriptions with at least one prespecified failure divided by the number evaluable under the composite indicator’s rules. The denominator for “failure by safety domain” was the number of prescriptions evaluable for that domain. “Number of safety-failure events” counted identifiable distinct events. One event could receive multiple domain labels, but was counted once in the event total; missing domains or outputs were not coded as zero and were to be reported separately. Denominator rules needed to be explicit; study counts, evaluator counts, or rating records could not substitute for prescription counts. The available data did not support cross-study pooled proportions.

Text similarity, the proportion of unique outputs, activity/dose consistency, generation-level score coefficients of variation (CVs), and inter-rater agreement were interpreted separately. When human studies did not report adverse events, this was recorded as “not reported,” not as no events or evidence of safety.

2.7 Methodological quality and risk of bias

Appraisal was matched to study design. The 32 Class A/B output-evaluation studies were assessed with a review-specific nine-domain framework covering clarity of the question and target, case/data source, model and generation conditions, prompt reproducibility, number of generations, evaluators/reference standards, outcome validity, unit of analysis, and safety-outcome boundaries. This framework is not a validated standardized risk-of-bias tool, and no total score was calculated.

The three Class C studies were described as randomized: W17 reports random assignment of 26 families (p.1 Abstract), W26 describes a randomized controlled study and random allocation (pp.1–2 Methods), and W27 reports that participants were randomly divided into two groups (pp.3–4 Methods). RoB 2 was therefore

retained for their randomized comparisons. The workbook contains 288 Class A/B framework judgments and 15 Class C RoB 2 domain judgments, each marked researcher-reviewed-confirmed in the saved source on 25 October 2026; reviewer identity is NR. No independent second reviewer or inter-rater statistic is recorded.

2.8 Evidence synthesis

We used a stratified narrative synthesis, with reporting informed by transparent synthesis principles for reviews without meta-analysis [3]. Classes A, B, and C were described separately. Within each study, model, prompt, comparator, scale direction, denominator, and time point were retained. We did not pool overall effects across different designs or scales, or rank models using scores from unrelated studies. Human outcomes were distinguished by within- versus between-group analysis, means versus medians, and combined-intervention context. Studies with missing full text were not included in verified-results narratives, although existing evidence leads were retained.

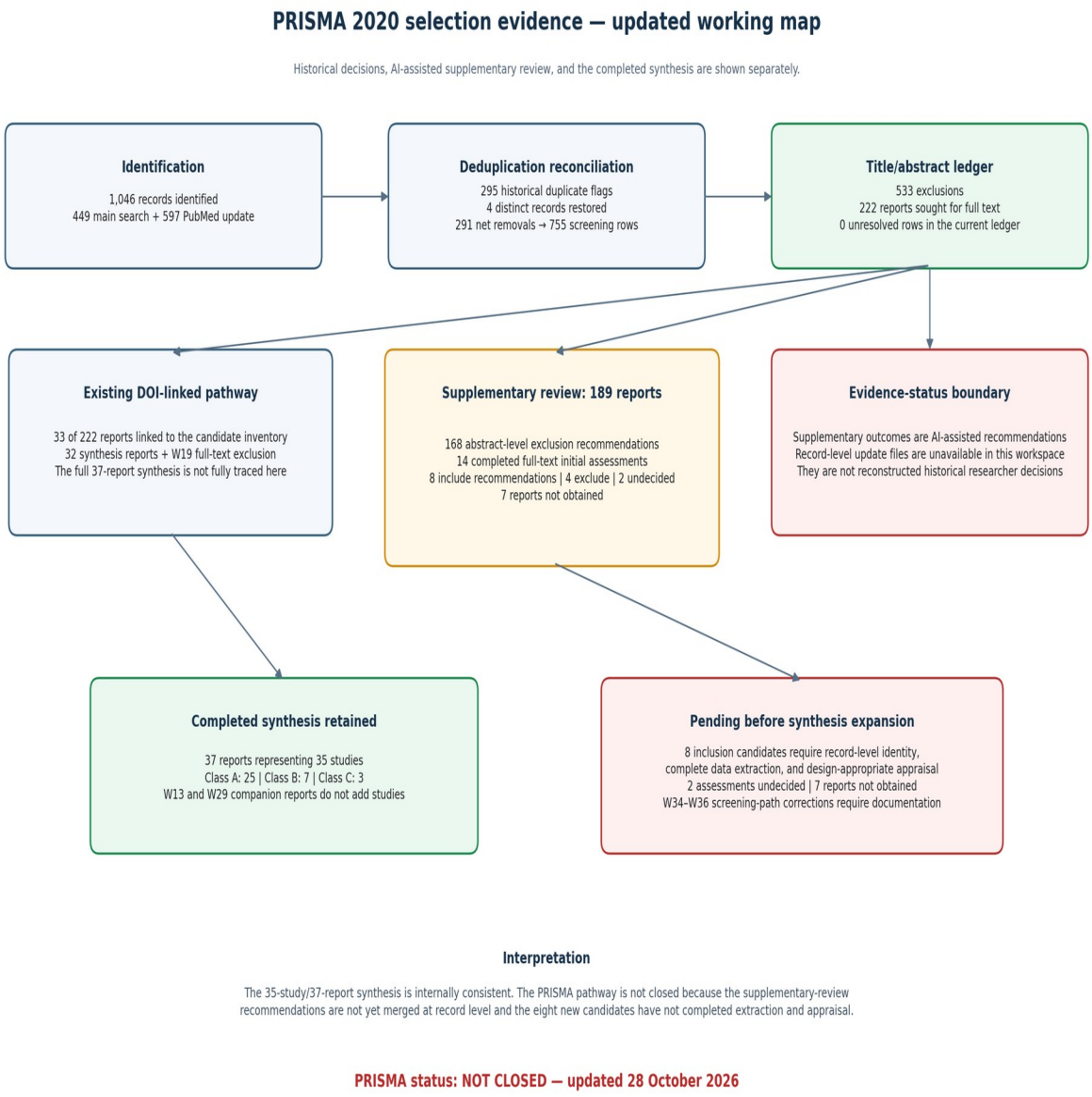
3. Results

3.1 Search results and study selection

The searches identified 1,046 records. The historical workbook flagged 295 duplicates; four distinct records were restored, giving 291 net removals and 755 title/abstract screening rows. The current ledger records 533 exclusions and 222 reports sought for full text. Thirty-three reports have exact DOI links to the existing candidate inventory. The supplementary review of the other 189 records yielded 168 abstract-level exclusion recommendations, 14 completed full-text initial assessments, and 7 reports not obtained. The 14 assessments comprised 8 inclusion recommendations, 4 exclusion recommendations, and 2 undecided reports. Because the record-level supplementary-review files and complete extraction/appraisal for the eight candidates were unavailable in this workspace, the completed synthesis remains 37 reports representing 35 studies (A=25, B=7, C=3). W19 remains the documented full-text exclusion, and the PRISMA flow remains open.

The 84 historical conflict rows retain their saved screening outcome and separate AI recommendation; row-specific reconciliation evidence remains incomplete. MAIN-0271, MAIN-0282, and MAIN-0290 have explicit current decisions dated 27 October 2026. MAIN-0076 carries an author-stated provisional preference toward exclusion, not a final eligibility decision. The second undecided full-text report and the seven unobtained reports cannot be identified from the aggregate update available here. W34–W36 remain in the synthesis set but require a documented correction or reinstatement pathway.

Figure 1. PRISMA 2020 selection evidence map. The 28 October 2026 supplementary-review status is shown separately from historical researcher decisions; unresolved nodes remain explicitly open.



The 751 historical working set and the 755-row screening inventory represent different stages: four previously duplicate-flagged records were restored. The current ledger records 533 exclusions and 222 full-text requests. Of the 222, 33 link by DOI to the existing candidate inventory and 189 entered supplementary review. The 189-record update comprises 168 abstract-level exclusion recommendations, 14 completed full-text initial assessments, and 7 unobtained reports. Recommendation status is not equivalent to a historical or final researcher decision.

3.2 Characteristics of the studies

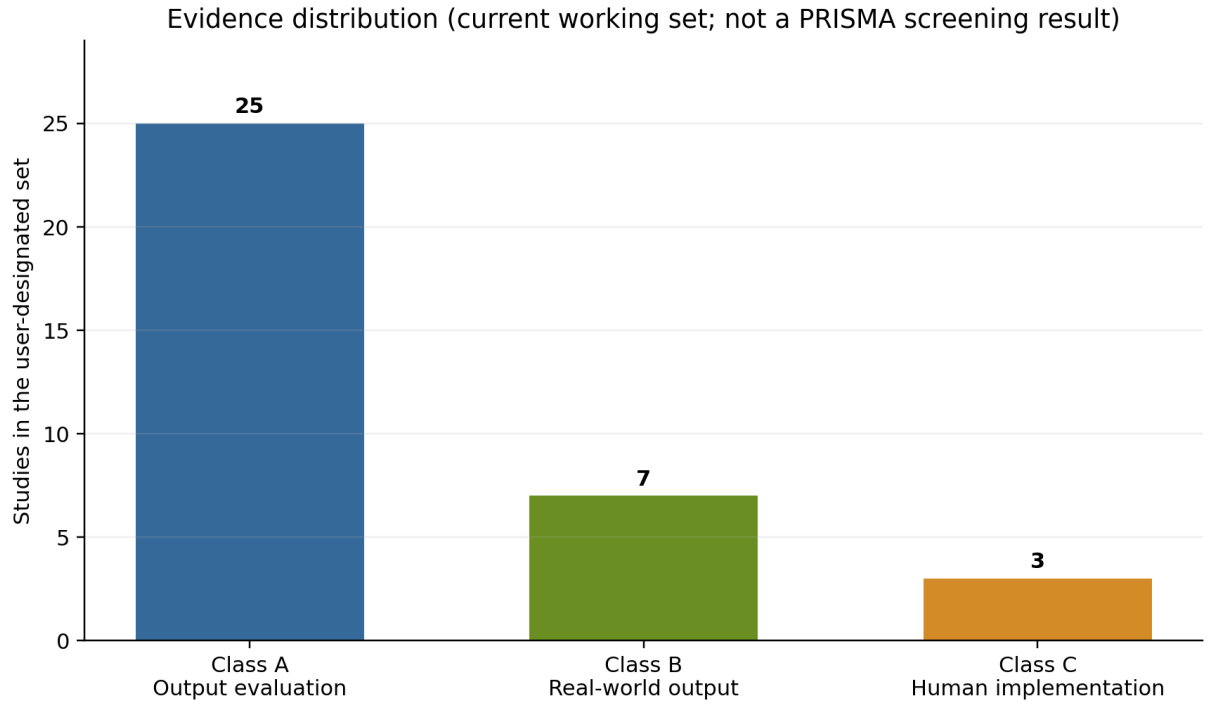
The current author-designated set contains 35 studies: 25 Class A, 7 Class B, and 3 Class C. Class A primarily evaluated prescription completeness, quality, safety expression, and reproducibility in simulated or standardized scenarios. Class B used real user or patient data, or real interactions, to generate and evaluate

outputs. Class C involved implementation of LLM-generated plans in people. Results from the three classes were not combined.

Units of analysis in output-evaluation studies included cases, generated texts, models, evaluators, and rating records; these were not added into a single sample size. Real-world data in Class B did not imply human implementation. For example, W03 used historical user logs for system training and offline evaluation, whereas W24 and W33 generated prescriptions from patient data without verifying actual patient implementation.

Author, year, DOI, design, model, exercise context, evaluation method, and evidence location for the 35 studies are provided in the accompanying tables. W13 and W29 each count as one study, with their linked preprints retained as reports. W19 was not included in the synthesis.

Figure 2. Study-class distribution in the author-designated working set (A=25, B=7, C=3); these counts do not represent a completed screening flow.



Classes A, B, and C were synthesized separately; no overall effect was pooled across classes.

3.3 Completeness and quality of LLM-generated exercise prescriptions

Across the 35 studies, LLMs generally produced at least some prescription elements, such as activity, frequency, intensity, time, sets, or progression, but completeness and suitability varied. Class A studies covered resistance and running training, swimming, soccer, cardiac rehabilitation, musculoskeletal rehabilitation, diabetes, and kidney disease. Most used expert-developed scales, FITT/FITT-VP criteria, or guideline-adherence assessments; differences in scales and denominators precluded comparison of total scores across studies.

Additional full-text evidence supports a “generatable but requiring review” pattern. W08 generated 30-day FITT-VP plans for five simulated cardiac rehabilitation cases. Experts identified gaps in examination reminders, contraindications, and intensity control in nine diabetes plans in W10. W15, W16, and W20 found

that structural completeness could coexist with limited clinical specificity in high-risk or rehabilitation scenarios.

W22 coded recommendations for 26 population groups by frequency, intensity, time, type, volume, and progression and found limited overall completeness. The eight-week clinical scenarios in W25 specified activities and doses, but experts noted overly rapid progression. W30 and W34 compared soccer and marathon plans, respectively; multiple models generated structured periodized plans, but individual health screening and dynamic adjustment were limited.

Output evaluations using real-world data provided evidence closer to practical use. W11, W14, W23, W24, and W33 used real patient data or clinical prescriptions as inputs to compare individualization, workload, guideline adherence, or expert plans. However, use of patient data did not mean that patients implemented the plans, and output consistency could not be interpreted as a clinical effect.

W36's PhysioAgent used MedGemma, multi-agent orchestration, and a rule-based safety state machine to generate structured home exercise plans and was evaluated in 50 clinical scenarios and 200 simulated 30-day cycles. Transfer of real logs primarily validated the monitoring module and did not establish automated re-prescribing to real patients. System performance and safety-escalation rates remained offline or simulated evidence.

Specific activities, doses, and expert ratings from W01–W07, W21, W31, and W35 were retained in the supplementary tables. "Actionable" meant only that an output contained an activity and at least one dose dimension; it did not mean that the dose was appropriate for a particular patient.

Model, prompt detail, and language could affect prescription structure and ratings. Within-study comparisons can identify task-specific differences, but variation in model version, scenario, prompt, and scale precluded a cross-study ranking.

Full study characteristics and item-level evidence are provided in Tables 1 and 2. Information not reported in the source is coded as NR.

3.4 Guideline adherence and safety

Safety evidence primarily concerned output content. W08 reported expert safety ratings; W10 identified substantial gaps in plans for diabetes with comorbidities; W15 compared warnings and guideline adherence for at-risk populations; W16 assessed safety and risk awareness; W22 quantified errors such as missing medical-examination screening; W25 noted overly rapid progression; and W31 documented expert judgments that specific responses were unsafe. None of these represents a human injury event.

W13 coded contraindications, precautions, monitoring, and warnings as binary safety expressions. All three models had near-ceiling safety-expression scores. The authors noted that safety language did not guarantee dose completeness; Claude Sonnet 4.6 contributed to both generation and evaluation, raising a potential self-evaluation concern [23].

Class B studies W24 and W33 compared outputs based on real patient data with clinical plans and identified differences in prescription decisions. The rest-day and total-volume issues in W29 reflected structural plan risks. Real-data evaluation did not constitute human-implementation safety evidence.

A cross-study safety-failure rate could not be calculated because studies used different safety domains, output units, evaluator counts, and denominators. The three Class C studies did not adequately report human adverse-event outcomes; these were coded as NR, not as zero events.

3.5 Individualization and clinical applicability

The seven Class B studies were W03, W11, W14, W23, W24, W29, and W33. They generated and evaluated outputs using real logs, user needs, patient data, or clinical prescriptions; the use of “real-world data” alone did not make a study a human implementation study.

W03 and W29 provided system-performance and interaction evidence. W11 compared prescriptions generated by ChatGPT and an intelligent health-promotion system for patients with hypertension and comorbidities. W14 compared experts, novices assisted by AI, and AI-only protocols. W23 compared 16-week plans developed by GPT-4 and a coach for the same real individual. W24 and W33 compared prescriptions using data from patients with knee osteoarthritis and stable angina, respectively.

These studies show that LLMs can generate structured plans from real-world inputs, but outcomes were primarily expert ratings, algorithmic proxy measures, or output consistency. Adherence, effectiveness, or harm cannot be inferred unless plans are implemented and outcomes are observed.

Individualization should be assessed across input information, activity selection, dose, progression, contraindications, and resource constraints—not solely by output length or whether personal information is mentioned.

3.6 Differences by model and prompting approach

Model-comparison results were task-dependent. W09, W18, W21, W30, W32, and W35 indicated that model version, prompt detail, or language could affect outputs. Differences in scales, number of generations, and evaluators precluded identification of a universally superior model.

More detailed prompts generally improved structural completeness but did not reliably resolve clinical screening, intensity matching, or dynamic adjustment. The included studies consistently retained professional review as a boundary condition.

3.7 Stability across repeated generations

W13 provided the most extensive repeated-generation evidence: GPT-4.1, Claude Sonnet 4.6, and Gemini 2.5 Flash each generated 20 outputs for six clinical scenarios, for 360 outputs in total. Mean semantic similarity was 0.955 for GPT-4.1, 0.903 for Claude, and 0.950 for Gemini. All GPT-4.1 outputs were unique, whereas only 27.5% of Gemini outputs were unique. High similarity may reflect stable semantics or text repetition [23].

W18 and W21 further indicated that input granularity, model, and task affected rating variability. The small number of repeated generations in W05 was insufficient to establish equivalence. Stability should be evaluated separately for text, activities, doses, and risk-management decisions.

Safety expressions in W13 were near ceiling, although some resistance-intensity outputs could not be classified. This illustrates that consistent safety wording does not establish prescription completeness.

Similarity, proportion of unique outputs, rating CVs, and expert agreement were not pooled across studies because they measure different constructs.

3.8 Human implementation studies

All three Class C studies implemented exercise plans generated with LLM involvement (Table 3). Sample sizes and supervision conditions differed, so outcomes were not pooled.

W17 compared four weeks of physical activity across 26 families of children with autism spectrum disorder (ASD), with 13 families per group [12]. The report identified ChatGPT 4.0. Recommended activities included five minutes of balance walking and ten minutes of catch-and-throw. Parents received three 40-minute training sessions and, with researcher support, implemented the plan three times weekly for 40 minutes per session.

Mean LTEQ scores increased from 6.69 to 34 in the intervention group and changed from 7.46 to 7.23 in the control group; the Table 3 interaction was $F(1,24)=353.069$, $P<0.001$, $\eta p^2=0.936$. Outcomes were parent-reported and occurred alongside training, parent–child participation, and WhatsApp support; the change cannot be attributed to the LLM alone.

W26 provided an eight-week exercise program to nine healthy overweight men aged 19 years (six intervention, three control) [14]. The Methods summarized aerobic, resistance, and flexibility training for 45–60 minutes three times per week, whereas the appendix described 20–30-minute HIIT sessions in weeks 3–4. The prescription examples and summary differed, and session-level completion was not reported. At week 8, median BMI was 22.99 versus 24.46 kg/m² ($P=0.020$), heart rate after a 3-minute in-place knee-raise test was 128 versus 152 bpm ($P=0.019$), and 30-second sit-to-stand counts were 37 versus 32 ($P=0.019$). These were between-group endpoint medians, not means or between-group changes. The control group received no exercise intervention; results do not show that LLM plans outperform professional prescriptions. The very small sample and an injury-related exclusion criterion further limit interpretation.

W27 compared a fixed GPT-4 plan with a personal-trainer plan in two groups of 10 men, training for 12 weeks, three times per week, for 90 minutes per session [15]. In the GPT group, left-arm strength increased from 20.31±3.80 to 22.99±4.05 kg (within-group $P<0.001$). Several measures changed within the trainer group, whose plan was reassessed and adjusted every four weeks; the GPT group's baseline maximal-strength prescription was not modified over 12 weeks. Support conditions therefore differed. A significant within-group result in one arm and a nonsignificant result in another do not establish a comparative effect.

None of the three studies reported human adverse-event outcomes; these were coded as NR. Non-reporting does not mean zero events or establish implementation safety. W26 listed injuries during training as an exclusion criterion, but the available report did not establish how many participants were affected; injury or withdrawal counts were not inferred.

3.9 Methodological quality and risk of bias

The 32 Class A/B studies were assessed with a review-specific nine-domain framework developed for this review; it is not a validated standardized risk-of-bias tool, and no overall score or cross-study low-risk proportion was calculated. The three Class C studies report randomized allocation and were assessed with RoB 2. The saved appraisal results contain 288 Class A/B and 15 Class C domain judgments, all marked researcher-reviewed-confirmed on 25 October 2026; reviewer identity is NR and no independent second reviewer is claimed. In W13, Claude Sonnet 4.6 generated outputs and assisted with FITT classification and safety-expression identification, creating a potential model self-evaluation/measurement-bias concern.

Across the three Class C studies, the saved researcher-reviewed RoB 2 judgments identify at least one high-risk domain for each target outcome. The principal concerns were parent-reported, unblinded outcomes in W17; possible post-randomization exclusion and selective reporting in W26; and reliance on within-group significance without adequate between-group effects in W27. These judgments address bias in the reported human outcomes; they do not establish whether the LLM itself caused the observed changes.

4. Discussion

4.1 The capabilities and limits of structured prescription generation

The current author-designated set indicates that LLMs can generate structured plans across exercise and rehabilitation contexts. Their practical value lies primarily in rapidly producing drafts for professionals to review and revise. The number of studies has not eliminated limitations in dose, individual fit, risk management, or dynamic adjustment.

Evidence remained weighted toward output evaluation: 25 Class A studies and 7 Class B studies, compared with only 3 Class C studies involving human implementation. This distribution suggests that the field is better positioned to demonstrate that plans can be generated than to establish long-term effectiveness, safety, or superiority to professional practice.

4.2 Dose, individualization, and risk management

A complete FITT-VP structure does not guarantee an appropriate dose. In W06, more detailed plans received higher overall ratings, yet health screening remained only moderate and progression concerns persisted. The MRC 2/5 scenario in W31 suggests that an apparently reasonable activity may overlook bilateral weakness, feasibility of assistance, or disease-specific evidence. Findings should be interpreted in relation to the specific activity and patient context; an overall score cannot replace professional judgment.

Risk-management evaluation also requires an explicit reference standard. Experts may disagree about the same frequency or treatment approach, as illustrated by W31; time constraints in W29 may also make it difficult to satisfy rest-day rules. Future assessments should record the model recommendation, reference decision, rationale, and uncertainty separately. Expert disagreement or one missing safety statement cannot be used to calculate a general hazard rate.

More detailed prompts may help but also increase the user burden. In W06, prompt detail and the number of follow-up queries changed together; W05 also had limitations in rater agreement. These findings do not support advising all users that a “better prompt” alone will produce a reliable prescription. Models may help structure information, but professionals must still determine whether the input is sufficient for prescribing.

4.3 Stability is distinct from prescription quality

W21 provides a clear example: the model with lower mean ratings had a smaller generation-level CV. Repeatedly generating similar outputs may indicate stability, but it may also reproduce the same defect. In W05, the absence of statistically significant rating differences was limited by few repetitions and inadequate rater agreement; it should not be interpreted as evidence of equivalence.

Future evaluations should report reproducibility separately for text, activities, doses, and risk-management decisions. Model parameters, conversation history, and version should be documented for identical inputs. Rating stability should be assessed alongside content-level consistency. Better performance in one study does not establish superiority across populations or later model versions, and this review does not support a universal ranking.

4.4 The evidence gap between output safety and human safety

Output-safety scores can identify content requiring professional review, but they do not directly predict injury. The 4/12 in W31 represents evaluators, the 4.8/5 in W03 represents a plan rating, and the rest-day rule in W29 concerns plan structure. These measures have different denominators and targets and should not be converted into a common “safety rate.”

None of the three human studies reported adverse events, so it is unclear whether no events occurred or events were not systematically monitored. Human studies should prespecify event type, severity, exposure time, withdrawal reason, and supervision intensity and report all participants. Exclusion after injury warrants particular transparency to avoid underestimating risk through complete-case analysis. The evidence is insufficient to conclude either that LLMs are generally safe or that all output problems translate into actual harm.

4.5 Real-world application and professional collaboration

Real data do not automatically establish real-world effectiveness. W03 used real logs, but satisfaction and goal matching included proxy measures; W29 used actual user needs but adjusted plans in simulated follow-up scenarios. These studies inform system development and interaction design but do not replace long-term follow-up.

Class C studies show that LLM-generated plans can be implemented when combined with parent training, researcher support, or coach involvement. In exercise science and rehabilitation, a more plausible role is to organize drafts, support communication, and document revisions—not to treat generated text as an individualized clinical decision. For people with disease, marked muscle weakness, or higher loading risk, professionals should review contraindications, intensity, progression, and stopping criteria and then adjust the plan based on observed responses. The current evidence does not show that this collaborative model is superior to usual professional care.

As background, WMRE2030 proposes a daily-scale closed loop integrating wearable devices, multi-omics, and real-time feedback [22]. This framework highlights the value of placing prescriptions within ongoing observation and adjustment rather than evaluating a single text output. However, it is a framework review, not a clinical trial eligible for this review, and cannot be used to imply that feedback procedures or effects were observed in the included studies. The FITT-VP and swimming studies in the learning materials are already represented as W07 and W21 and were not counted again.

4.6 Heterogeneity and limitations of this review

The title/abstract ledger contains a current or recovered decision for all 755 rows: 533 exclusions and 222 requests for full text. A supplementary review of 189 previously unlinked reports reduced the uncharacterized queue to 168 abstract-level exclusion recommendations, 14 completed full-text initial assessments, and 7 unobtained reports. The 14 assessments yielded 8 inclusion recommendations, 4 exclusion recommendations, and 2 undecided reports. However, the record-level evidence files for this update were not available in the current workspace, so these recommendations could not be merged into the formal screening ledger, data extraction, or quality appraisal. W34–W36 also retain an undocumented cross-stage discrepancy. These issues prevent a fully closed PRISMA flow.

Second, complete original exports or execution records remain unavailable for some databases. The 295 records are historical duplicate labels; the net removal of 291 reflects the restoration of four distinct reports. The values 751 and 755 represent different working inventories before and after this correction.

The Class A/B nine-domain framework was developed for this review and has not been externally validated. The saved quality results contain 303 item-level judgments (288 A/B and 15 C) marked researcher-reviewed-confirmed on 25 October 2026. The reviewer is not named in the source, and no independent second reviewer or agreement statistic is recorded. RoB 2 was applied to the three Class C studies, each of which reports randomized allocation. Output-safety assessments remain distinct from human adverse events.

Finally, models, versions, prompts, scenarios, scales, and units of analysis were highly heterogeneous, and reporting of human adverse events was limited. We therefore conducted no meta-analysis and calculated no overall safety rate or model ranking.

4.7 Future research

Future output studies should publish original prompts, parameters, all generated outputs, and failed outputs. Standardized cases should cover different training levels, conditions, and actionable constraints, with prespecified reference standards for risk management. Repeated generations should be sufficient to examine dose, activity, and safety decisions separately, rather than reporting only similarity or mean ratings.

Real-world studies should distinguish contributions from the model, knowledge constraints, professional support, and dynamic adjustment and should prespecify primary outcomes and units of analysis. Human studies need comparative designs that assess incremental value and should report adherence, follow-up, adverse events, and resource use. Appraisal tools should match the study design rather than being selected for convenience.

5. Conclusion

The completed synthesis of 35 studies indicates that LLMs can generate structured exercise-prescription drafts, but completeness, dose suitability, individualization, safety expression, and reproducibility vary. Class A/B output evaluations and Class C implementation evidence are not interchangeable; 'adverse events not reported' does not mean that no events occurred. The evidence supports supervised use as an assistive tool, not independent prescribing. The supplementary review identified eight additional inclusion candidates, but these have not been incorporated because complete extraction and quality appraisal are not available. Two full-text assessments remain undecided, seven reports were not obtained, and W34–W36 require a documented screening-path correction. The manuscript therefore remains an internally consistent author-review draft rather than a submission-ready final review.

Supplementary tables

Tables describe the current author-designated 35-study synthesis set (25 Class A, 7 Class B, and 3 Class C). NR = not reported in the source; PENDING = unresolved evidence or researcher confirmation. Class A = standardized/simulated output evaluation; Class B = real-world data output evaluation; Class C = human implementation. PDF locations refer to physical page numbers.

Table 1. Characteristics of the author-designated study set (Part A).

Study ID	Author/year; class	Study design and evaluation	Population/source and LLM
W01	Dergaa 2024; A	Evaluation of LLM outputs for simulated cases; critical expert appraisal	GPT-4; five synthetic healthy/disease cases
W02	Havers 2025; A	Multimodel comparison of simulated training plans; expert assessment	GPT-3.5, Copilot, Gemini; fictional advanced strength trainee
W03	He 2026; B	Model development and performance evaluation using real-world training logs	LLM-SPTRec and general LLaMA-2; SportFit-1M user data
W04	Washif 2024; A	Comparison of simulated resistance-training prescriptions	GPT-3.5, GPT-4; two fictional healthy individuals
W05	Havers 2025; A	Repeated generation and expert assessment	GPT-4, Gemini; fictional hypertrophy trainee
W06	Düking 2024; A	Simulated running plans, prompt-information gradient, and expert assessment	ChatGPT; fictional runner
W07	Feng 2026; A	Multimodel FITT-VP output evaluation in 30 simulated cases	GPT-4o, Claude 3.7, DeepSeek R1, Grok-3; 30 synthetic cases
W08	Perrero 2026; A	Output evaluation in five simulated cardiac rehabilitation cases	DeepSeek-V3; five cardiac rehabilitation cases
W09	Genç 2026; A	Mixed-method expert evaluation of simulated resistance-training plans	ChatGPT-3.5, 4o, 4.1; fictional healthy young men
W10	Akrimi 2025; A	Expert evaluation of nine plans for three fictional type 2 diabetes cases	ChatGPT-4o; three type 2 diabetes cases

Study ID	Author/year; class	Study design and evaluation	Population/source and LLM
W11	Xu 2024; B	Comparison of prescriptions from two systems using real patient data	ChatGPT-4.0, IHPS; 10 patients with hypertension and comorbidities
W12	Wachholz 2025; A	Qualitative expert assessment of one AI running plan and user survey	ChatGPT-3.5; novice half-marathon runner scenario; 119 survey respondents
W13	Lee 2026; A	Repeated-generation study: 3 LLMs × 6 clinical scenarios × 20 generations per scenario (360 outputs)	GPT-4.1, Claude Sonnet 4.6, Gemini 2.5 Flash; six clinical/health scenarios
W14	Sortino 2026; B	Cross-sectional blinded expert evaluation based on five real orthopedic prescriptions	AI alone, novice plus AI, experts; five real orthopedic prescriptions
W15	Choi 2026; A	Structured expert assessment of complex high-risk simulated cases	Gemini 2.5; three high-risk synthetic cases
W16	Cheng 2026; A	Blinded expert assessment of two models in five virtual axillary web syndrome cases	ChatGPT, DeepSeek; five virtual axillary web syndrome cases
W17	Aydemir 2025; C	Controlled pre–post feasibility study with human implementation	ChatGPT; 26 families of children with ASD
W18	Yang 2026; A	Crossover repeated-generation design with blinded expert assessment	GPT-4; healthy training scenario

Table 1 (continued). Characteristics of the author-designated study set (Part B).

Study ID	Author/year; class	Study design and evaluation	Population/source and LLM
W20	Cali 2026; A	Mixed-method expert assessment in five simulated sports-injury cases	ChatGPT-4.1; five sports-injury cases
W21	Zhao 2026; A	Expert evaluation of repeated multimodel generation in a simulated swimming scenario	ChatGPT, Gemini, DeepSeek, Doubao; one simulated 100-m freestyle swimmer
W22	Zaleski 2024; A	Mixed-method coding of outputs for 26 standardized populations	ChatGPT; 26 clinical populations/scenarios
W23	Li 2025; B	Comparison of plans generated from one real individual's data	GPT-4; one real individual
W24	Gürses 2025; B	Output evaluation using data from 40 real patients with knee osteoarthritis	ChatGPT-4o, Gemini Advanced; 40 patients with knee osteoarthritis
W25	Onan 2025; A	Program generation and expert evaluation in three clinical case scenarios	ChatGPT-4.0; three clinical cases
W26	Philuek 2025; C	Pilot controlled study with human implementation	ChatGPT/GPT-4; healthy overweight young adults
W27	Genç 2025 (Digital coaches); C	Controlled human implementation study	GPT-4 (confirmed in DOI abstract); 20 men
W28	Enichen 2025; A	Blinded comparison of AI and expert recommendations in 30 fictional cases	GPT-3.5; 30 fictional patient cases
W29	PlanFitting 2025; B	Evaluation of an interactive exercise-planning system with real-user interaction	LLM conversational agent; user study of personalized exercise planning
W30	Yang 2026 (soccer); A	Expert comparison of soccer training plans generated by six models	Six generative AI systems; soccer-training scenarios

Study ID	Author/year; class	Study design and evaluation	Population/source and LLM
W31	Schütze 2026; A	Expert evaluation of six standardized inclusion body myositis questions	ChatGPT; exercise scenarios for people with inclusion body myositis
W32	Fu 2026; A	Multimodel evaluation of rehabilitation program quality/readability across clinical scenarios	Four LLMs; multiple musculoskeletal scenarios
W33	Hüzmeli & Akkuş 2026; B	Output comparison using data from 40 real stable-angina cases	ChatGPT-5.2, Gemini Advanced; stable angina/CCS cases
W34	Montaruli 2026; A	Descriptive comparison of original marathon plans from eight AI engines	Multiple generative AI systems; beginner, intermediate, and advanced runners
W35	Bezruk 2026; A	Multimodel, multilingual output evaluation in simulated CKD cases	ChatGPT, Gemini, Copilot; CKD/hemodialysis rehabilitation scenarios
W36	PhysioAgent 2026; A	Multi-agent system evaluation in 50 clinical scenarios and 200 simulated follow-up cycles	Multi-agent LLM; 50 clinical scenarios and 200 simulated events

Table 2. Prescription quality, safety-related evidence, and principal findings.

Study ID	Principal finding	Safety evidence / denominator	Evidence location
W01	Generally safe and conservative; disease specificity and goal precision were limited.	Class A/B output-safety evidence only; Class C human adverse events NR.	Methods pp. 2–3; Table 2 p. 5; pp. 8–12
W02	Overall performance was moderate; model differences were limited, and professional revision remained necessary.	Output-safety evidence only; no human adverse-event inference.	Methods/Tables 4–5, pp. 2–7; outputs, pp. 16–26
W03	Safety score: 4.8/5 for the knowledge-graph system vs 3.1/5 for the general LLM.	Output-safety rating; not a human safety outcome.	Schema p. 3; Methods/Results pp. 6–11
W04	Outputs could serve as drafts; sex-specific differences were insufficiently addressed and plans required refinement.	Output-safety evidence only.	Methods/Results pp. 2–5
W05	Twenty-seven of 28 comparisons were not statistically significant; more detailed prompts produced higher quality.	Output evaluation; not a human adverse-event outcome.	pp. 5–6; appendices pp. 16–33
W06	Scores below 3 were reported for 19, 11, and 1 indicators, respectively; ratings improved with additional input.	Output-safety evidence only.	Methods pp. 2–5; outputs pp. 9–17
W07	Total scores were 50.23, 47.42, 44.02, and 40.30/60, respectively.	Output-safety evidence only.	Methods/Tables 2–3, pp. 3–6
W08	Guideline adherence was 4.5/5 and safety was 5.0/5; no clearly unsafe prescription was identified.	Expert-rated output safety; not human safety.	Abstract p. 1; Methods/Results/Table 2, pp. 2–5
W09	Version 4.1 received higher ratings; risk recognition and individualization remained limited.	Output-safety evidence only.	Abstract pp. 1, 3–5; Methods/output; Results pp. 6–13
W10	Most plans had safety gaps, most notably in complex comorbidity scenarios.	Expert assessment of generated plans; not human adverse events.	Abstract and Tables 2–4/Results, pp. 1, 3–8
W11	For GPT: accuracy 4.64/6, completeness 2.25/3, and applicability 3.05/4.	Output-safety evidence only.	Methods p. 1; prescriptions/results pp. 5–12

Study ID	Principal finding	Safety evidence / denominator	Evidence location
W12	Trust in the AI plan was rated 3.73 vs 3.36; this did not validate safety.	User trust is not a safety outcome.	Methods/Results pp. 1–4
W13	Semantic similarity ranged from 0.903 to 0.955; unique-output proportions ranged from 27.5% to 100%.	Safety-expression analysis of generated outputs; not human adverse events.	Abstract p. 2; Methods pp. 4–6; Tables 1–6 pp. 7–12; Supplement pp. 19–24
W14	Expert-generated plans rated highest, novice-plus-AI plans intermediate, and AI-only plans lowest.	Output comparison; no human implementation safety outcome.	Abstract p. 1; Methods/Results pp. 3–10
W15	Ratings generally improved at Stage 2 but were inconsistent at Stage 3; ICC=0.139.	Expert evaluation of output safety; not patient harm.	Abstract p. 1; Methods/Results and rubric
W16	DeepSeek was rated higher overall; ChatGPT more often showed generalization and differential-diagnosis omissions.	Output-level risk evaluation only.	Methods/Tables 2–5/Results pp. 4–9
W17	Intervention-group LTEQ increased from 6.69 to 34; adverse events were not reported.	Human implementation; adverse-event outcome NR.	Methods/Tables 2–3 pp. 3–6
W18	With detailed prompts, safety ratings were 3.95 vs 3.33/5 and output variability was lower.	Output safety rating; not human safety.	Abstract p. 1; Methods/Results
W20	Overall rating 3.85/5; ACL 1.88 and clavicle fracture 5.00.	Expert-rated output; not human adverse events.	Abstract p. 1; six-week program and expert assessment
W21	ChatGPT had the highest quality ratings; Doubao was most stable but had lower quality ratings.	Repeated-generation and expert-rated output evidence.	Methods/Tables 2 and 5, pp. 3–10
W22	Completeness and accuracy were reported; safety denominators cannot be inferred directly from 26 cases.	Full-text extraction of safety denominators remains incomplete in the source table.	Abstract p. 1; coding/results pp. 3–6
W23	Compared 16-week GPT-4 and coach plans for the same real individual; activities and doses were specified, but evidence was limited to one case.	Class B output applicability; not human implementation safety.	Abstract p. 1; Methods/prompts/rubric and Results
W24	Compared ChatGPT-4o and Gemini Advanced outputs with rehabilitation prescriptions using data from 40 patients with knee osteoarthritis.	Class B output safety and prescription differences; not human adverse events.	Abstract p. 1; Methods/Results for 40 cases
W25	Eight-week scenario plans included activities and doses; experts noted overly rapid early loading and progression.	Output dose/progression risk; not human adverse events.	Abstract p. 1; Appendix/Table 2 and eight-week plans
W26	Endpoint BMI: 22.99 vs 24.46 kg/m² (P=0.020); post-knee-raise heart rate: 128 vs 152 bpm (P=0.019); sit-to-stand: 37 vs 32 repetitions (P=0.019).	Human adverse events NR; injury was an exclusion criterion, but event counts were not reported.	Methods p. 2; Table 2 p. 4; Appendix pp. 9–11
W27	GPT-group left-arm strength increased from 20.31±3.80 to 22.99±4.05 kg (within-group P<0.001); within-group significance does not replace a between-group effect.	Human adverse events NR.	Methods/Tables 1, 3–4, pp. 3–6

Study ID	Principal finding	Safety evidence / denominator	Evidence location
W28	Generated individualized plans from personal data and had professionals evaluate them; this supports output generation, not human implementation effects.	Output quality and potential risk; not human adverse events.	Methods/Table 1/Results pp. 1–3
W29	Fifteen of 18 initial plans met the study's weekly-volume requirement; 14/18 included rest days. Mean FITT ratings were 5.67/4.28/5.06/3.89.	Experts noted fatigue/injury concerns with consecutive training; no human implementation events were assessed.	pp. 6, 8–9; Figure 1 p. 13; Tables 2–3
W30	Ratings and readability were reported; individualized health screening was limited.	Full-text safety details require extraction from the cited source; no human safety inference.	Methods/Results pp. 1–5
W31	For Q4, 4/12 evaluators rated the response incorrect and unsafe. For Q3, 3/12 rated it complete/accurate, 6/12 safe but incomplete, and 3/12 safe but basic.	Expert output-safety judgments; not four patients or four prescriptions with injury.	Methods/Table 1/Results pp. 2–6
W32	Mixed consultation tasks; only the actionable rehabilitation subset was extracted. The total of 1,476 responses is not a prescription denominator.	Output-safety evidence only; no human adverse-event inference.	Methods/quality/readability/results pp. 3–18
W33	Compared LLM-generated exercise rehabilitation plans using data from people with stable angina; this is real-data output evaluation.	Output quality and risk management; not human adverse events.	Methods/Tables/Results pp. 1–9
W34	Multiple models generated structured marathon periodization plans, but health screening, individualization, and dynamic adjustment were limited.	Output quality and potential loading risk; not human adverse events.	Prompts/Tables 1–3/comparison pp. 3–13
W35	English outputs generally described progression, monitoring, and intradialytic exercise in greater detail; no comparable human-effect estimate was provided.	Vascular access, contraindications, and symptom monitoring were evaluated as output-safety content.	Methods/Results/Figures pp. 3–12
W36	PhysioAgent was evaluated for prescription generation and monitoring across 50 clinical scenarios and 200 simulated 30-day cycles; transfer of real logs did not establish human re-prescribing.	Rule-based safety escalation and simulated monitoring; not human adverse events.	Architecture/Datasets/Results/ Ablation pp. 4–10

Table 3. Human implementation evidence.

Study ID	Design	Intervention and dose	Outcome and original analysis / safety and limitations
W17	Controlled pre–post feasibility study with human implementation	Four weeks; three sessions per week; 40 min per session.	Intervention-group LTEQ increased from 6.69 to 34; adverse-event outcome not reported (NR).
W26	Pilot controlled human implementation study	Eight weeks; six intervention and three control participants.	Endpoint BMI 22.99 vs 24.46 kg/m² (P=0.020); post-knee-raise heart rate 128 vs 152 bpm (P=0.019); sit-to-stand 37 vs 32 repetitions (P=0.019). Very small sample; timing and impact of injury-related exclusion unclear; adverse events NR.

Study ID	Design	Intervention and dose	Outcome and original analysis / safety and limitations
W27	Controlled human implementation study	Twelve weeks; two groups of 10 men; three sessions per week; 90 min per session.	GPT-group left-arm strength 20.31±3.80 to 22.99±4.05 kg (within-group P<0.001). Outcomes were primarily within-group; support conditions differed; adverse events NR.

Table 4. Structured appraisal evidence and confirmation status.

Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W01	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W02	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W03	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W04	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W05	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W06	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W07	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W08	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W09	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W10	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W11	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W12	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W13	A/B / review-specific nine-domain output appraisal (not externally validated)	Claude Sonnet 4.6 generated outputs and assisted classification; potential self-evaluation concern. Output safety is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

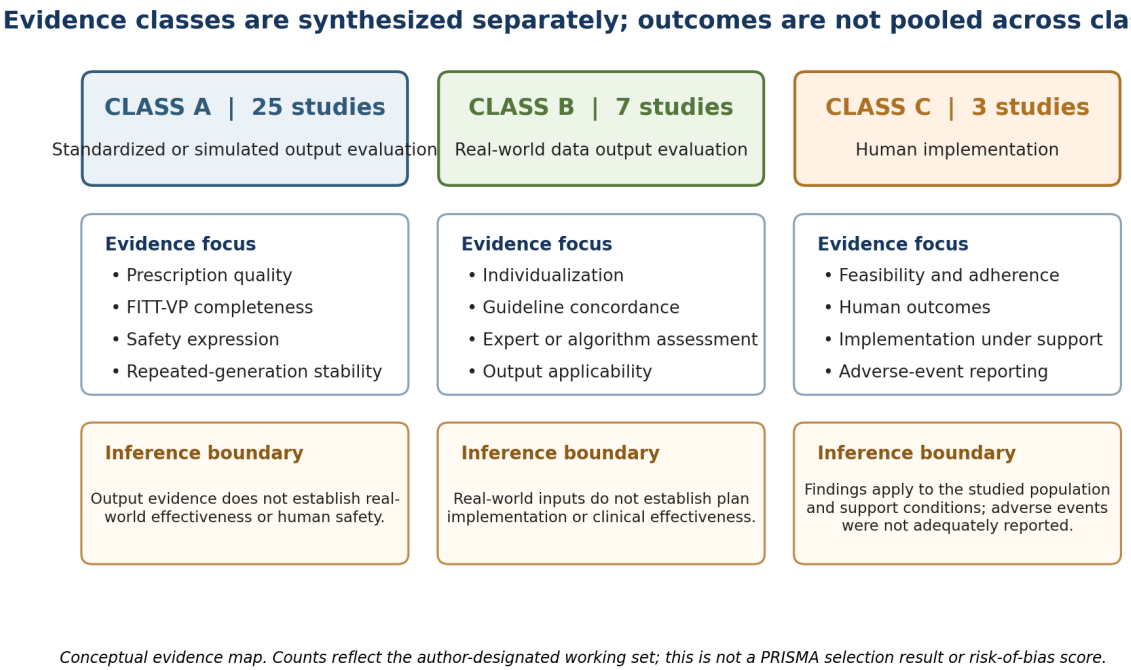
Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W14	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W15	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W16	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W17	C / RoB 2; randomized allocation reported in the source study	Five domain-level judgments with cited evidence; evaluates human implementation outcomes, not output safety.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W18	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W20	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W21	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W22	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W23	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W24	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W25	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W26	C / RoB 2; randomized allocation reported in the source study	Five domain-level judgments with cited evidence; evaluates human implementation outcomes, not output safety.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W27	C / RoB 2; randomized allocation reported in the source study	Five domain-level judgments with cited evidence; evaluates human implementation outcomes, not output safety.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W28	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W29	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W30	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W31	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W32	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W33	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W34	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.
W35	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

Study ID / class	Framework and target	Evidence / AI-assisted appraisal	Researcher confirmation status
W36	A/B / review-specific nine-domain output appraisal (not externally validated)	Domain-level judgments with cited evidence; no unvalidated overall rating calculated. Output-safety evidence is not a human adverse-event outcome.	Saved source status: researcher-reviewed-confirmed (25 Oct 2026); reviewer identity NR; no independent second reviewer.

Figure 3. Evidence classes and boundaries of inference.



Conceptual evidence map; it does not represent an effect size, risk-of-bias rating, or adverse-event incidence.

Declarations—author confirmation required

Data availability: PENDING: confirm whether the extracted study-level dataset, screening records, and appraisal workbook will be made publicly available and provide a repository link if applicable.

Author contributions: PENDING: authors must confirm roles using the journal's required taxonomy. No contribution statement is inferred from the supervisor field.

Funding: PENDING: author confirmation required; no funding source was supplied in the audited project materials.

Conflict of interest: PENDING: author confirmation required.

Ethics statement: Not applicable to this review of published reports, subject to author and journal confirmation.

Generative AI statement: AI tools assisted with record organization, evidence extraction, and language editing. The authors must confirm the exact tools, scope, and responsibility statement before submission.

Supplementary material: Supplementary extraction and evidence files are prepared in the accompanying workbook; final journal file naming and upload requirements remain to be checked against current author instructions.

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Appendix: Reports for the 35-study author-designated set (journal formatting to be confirmed)

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